

Using Elastic File System With Instances.

1. What is Amazon EFS

Amazon Elastic File System (EFS) is a **fully managed Network File System (NFS)** provided by **Amazon Web Services** that allows **multiple EC2 instances** to **access the same storage simultaneously**.

Instead of attaching storage to just one instance (like EBS), EFS acts like **shared network storage**.

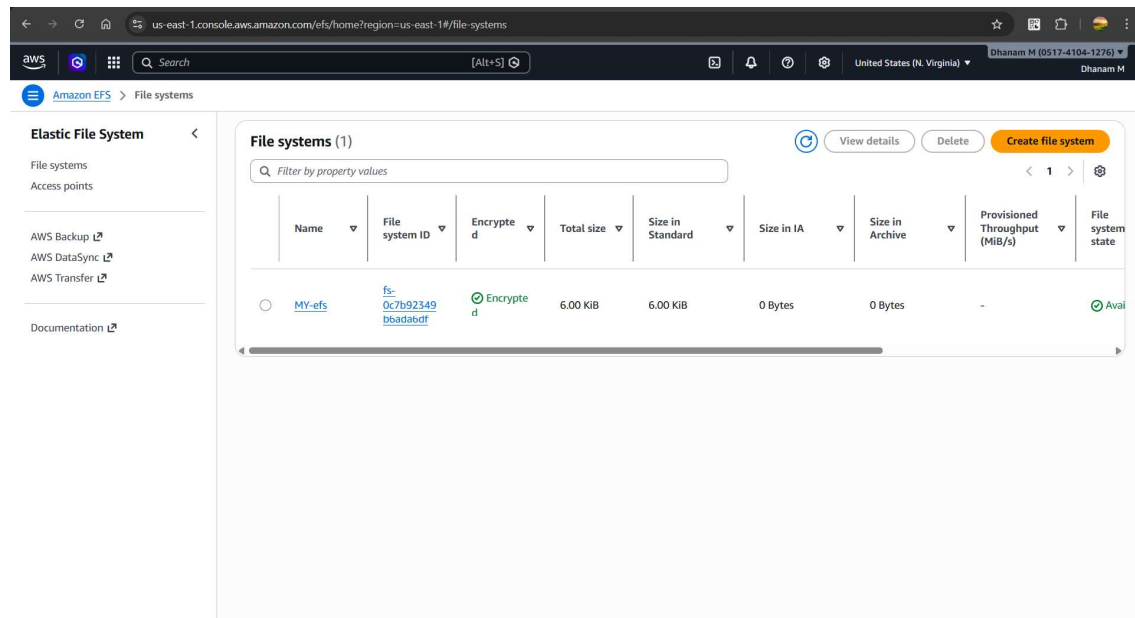
Key Idea

- One file system
- Many EC2 instances
- All read/write the same files

This is commonly used in **web servers, container workloads, and shared application data**.

I have created 3 instances and an EFS to connect with.

EFS



us-east-1.console.aws.amazon.com/efs/home?region=us-east-1#/file-systems/fs-0c7b92349b6ada6df

Amazon EFS > File systems > fs-0c7b92349b6ada6df

Elastic File System

File systems
Access points

AWS Backup [↗](#)
AWS DataSync [↗](#)
AWS Transfer [↗](#)

Documentation [↗](#)

MY-efs (fs-0c7b92349b6ada6df)

[Delete](#) [Attach](#)

General

[Edit](#)

Amazon resource name (ARN)
`arn:aws:elasticfilesystem:us-east-1:051741041276:file-system/fs-0c7b92349b6ada6df`

Performance mode
General Purpose

Throughput mode
Elastic

Lifecycle management
Transition into Infrequent Access (IA): 30 day(s) since last access
Transition into Archive: 90 day(s) since last access
Transition into Standard: None

Availability zone
Regional

Automatic backups
[Enabled](#)

Encrypted
473ea503-3820-448f-8848-0a5de921a784 (aws/elasticfilesystem)

File system state
[Available](#)

DNS name
`fs-0c7b92349b6ada6df.efs.us-east-1.amazonaws.com`

Replication overwrite protection
[Enabled](#)

[Metered size](#) | [Monitoring](#) | [Tags](#) | [File system policy](#) | [Access points](#) | [Network](#) | [Replication](#)

Metered size

Total size
6.00 KiB

us-east-1.console.aws.amazon.com/efs/home?region=us-east-1#/file-systems/fs-0c7b92349b6ada6df/network

Amazon EFS > File systems > fs-0c7b92349b6ada6df

Elastic File System

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Network

[Manage](#)

Availability zone (AZ-ID)	Mount target ID	Subnet ID	VPC ID	Mount target state	IPv4 address	IPv6 address	Network interface ID	Security groups
us-east-1a (use1-az1)	fsmt-0a7a7f4df95ebacf1	subnet-0618e0bb75502f409	vpc-0be33b50250021280 ↗	Available	172.31.4.7	N/A	eni-07f1f80eff80df715	sg-0a91bfc93233c9191 (default)
us-east-1b (use1-az2)	fsmt-003f7810e005601ca	subnet-0ca4939a51840d7c4	vpc-0be33b50250021280 ↗	Available	172.31.92.104	N/A	eni-09e20eb531b3249b7	sg-0a91bfc93233c9191 (default)
us-east-1c (use1-az4)	fsmt-067c6113c9480b064	subnet-07cb53f3c36994bc1	vpc-0be33b50250021280 ↗	Available	172.31.31.138	N/A	eni-0af23f5bd546468d4	sg-0a91bfc93233c9191 (default)
us-east-1d (use1-az6)	fsmt-0d2aeec6931602c91	subnet-013483e4615ac32f	vpc-0be33b50250021280 ↗	Available	172.31.44.104	N/A	eni-049666cbcf75bd35c1	sg-0a91bfc93233c9191 (default)
us-east-1f (use1-az5)	fsmt-0861e8ff5aeb56195a	subnet-07e40a724fb c9e7ed	vpc-0be33b50250021280 ↗	Available	172.31.78.128	N/A	eni-0c7364f145292dea0	sg-0a91bfc93233c9191 (default)

Instances

The screenshot shows the AWS Management Console for the 'us-east-1' region. The left sidebar contains navigation links for EC2, including Dashboard, AWS Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, and Capacity Manager. The main content area displays the 'Instances' page with a table of three running Linux instances. The instance i-0a03fbdad8e559505 is selected, and its details are shown below the table. The details include the instance ID, public IPv4 address (18.210.12.241), private IPv4 address (172.31.2.221), and public DNS (ec2-18-210-12-241.compute-1.amazonaws.com).

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
linux	i-0a03fbdad8e559505	Running	t3.micro	Initializing	View alarms +	us-east-1a	ec2-18-210-
linux	i-0f632027357561b91	Running	t3.micro	Initializing	View alarms +	us-east-1a	ec2-100-54
linux	i-07b24d9eb218620db	Running	t3.micro	Initializing	View alarms +	us-east-1a	ec2-3-235-

i-0a03fbdad8e559505 (linux)

Instance summary

Instance ID: i-0a03fbdad8e559505

Public IPv4 address: 18.210.12.241 | open address

Private IPv4 address: 172.31.2.221

Instance state: Running

Public DNS: ec2-18-210-12-241.compute-1.amazonaws.com | open address

Then we have to make sure that the both for EFS and EC2's Security Group must have to allow the NFS port.

For Instance.

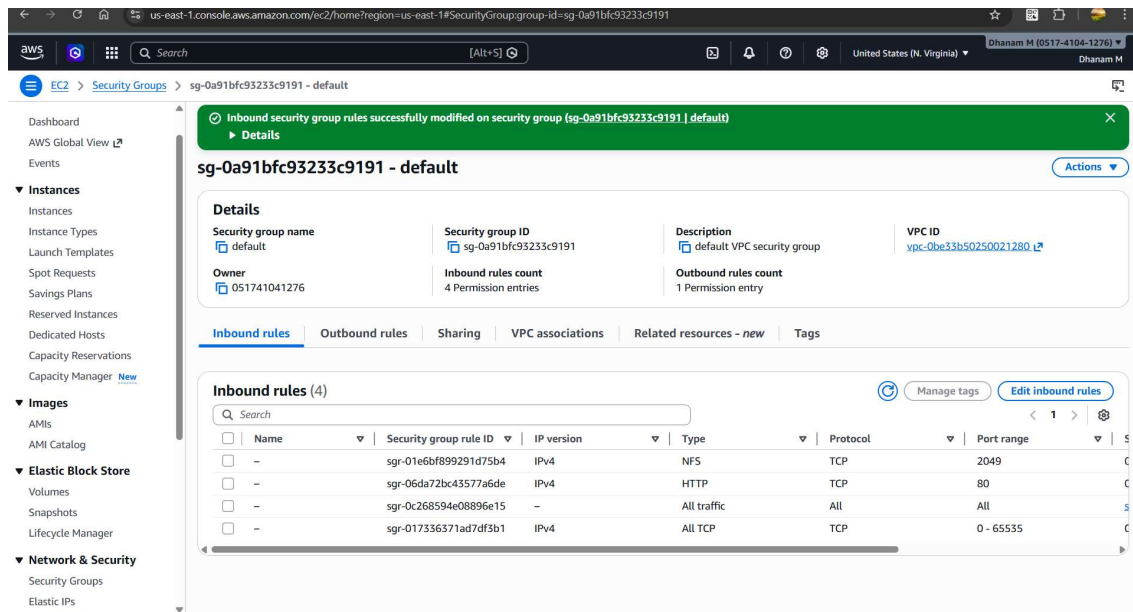
The screenshot shows the 'Edit inbound rules' page for a security group in the AWS Management Console. The page displays two inbound rules: one for SSH (port 22) and one for NFS (port 2049). The NFS rule is highlighted, showing its source as 'Anywhere' and its port range as 2049. The page also includes a warning message about rules with source 0.0.0.0/0 or ::0, and buttons for 'Cancel', 'Preview changes', and 'Save rules'.

Security group rule ID	Type	Protocol	Port range	Source	Description - optional
sgr-017f21aee695501eb	SSH	TCP	22	Custom	
-	NFS	TCP	2049	Anywhere	

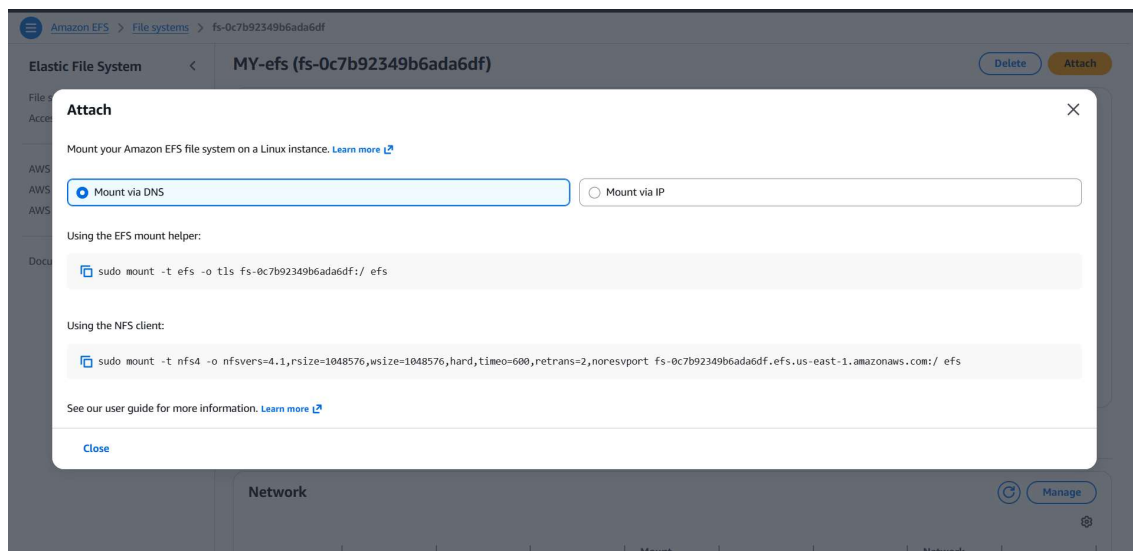
Rules with source of 0.0.0.0/0 or ::0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel Preview changes Save rules

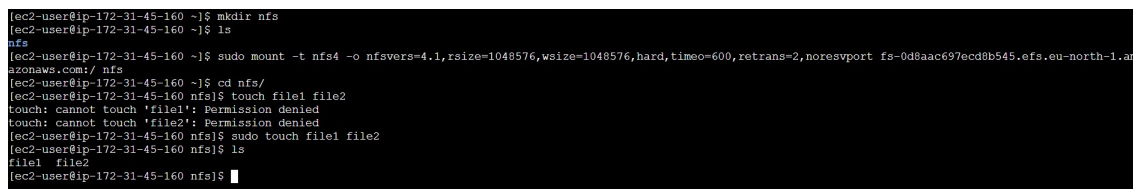
For EFS.



Then we have to copy the command given in the Attach EFS option.



We have to change it with our created file name.



Then I have do the same thing in the another instance.

And I have got the file that have been created in the main instance to another instance.

```
/m/^\nLast login: Wed Mar 11 11:20:24 2026 from 18.206.107.29\n(ec2-user@ip-172-31-2-221 ~)$ cd hari\n(ec2-user@ip-172-31-2-221 hari)$ touch file1\ntouch: cannot touch 'file1': Permission denied\n(ec2-user@ip-172-31-2-221 hari)$ sudo su\n(root@ip-172-31-2-221 hari)# touch file1\n(root@ip-172-31-2-221 hari)# ls\nfile1\n(root@ip-172-31-2-221 hari)# █
```

i-0a03fbdad8e559505 (linux) ✕

PublicIPs: 18.210.12.241 PrivateIPs: 172.31.2.221

```
/m/^\nLast login: Wed Mar 11 11:21:55 2026 from 18.206.107.27\n(ec2-user@ip-172-31-15-153 ~)$ sudo su\n(root@ip-172-31-15-153 ec2-user)# cd rahul\n(root@ip-172-31-15-153 rahul)# ls\nfile1\n(root@ip-172-31-15-153 rahul)# █
```

i-0f632027357561b91 (linux 2) ✕

PublicIPs: 100.54.230.253 PrivateIPs: 172.31.15.153

```
/m/^\nLast login: Wed Mar 11 11:21:22 2026 from 18.206.107.28\n(ec2-user@ip-172-31-9-187 ~)$ sudo su\n(root@ip-172-31-9-187 ec2-user)# mkdir mouri\n(root@ip-172-31-9-187 ec2-user)# sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsz=1048576,hard,timeo=600,retrans=2,noresvport fs-0c7b92349b6ada6df.efs.us-east-1.amazonaws.com:/ mouri\n(root@ip-172-31-9-187 ec2-user)# cd mouri\n(root@ip-172-31-9-187 mouri)# ls\nfile1\n(root@ip-172-31-9-187 mouri)# █
```

i-07b24d9eb218620db (linux 3) ✕

PublicIPs: 3.235.52.72 PrivateIPs: 172.31.9.187